Homework 2

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Exercise 1

$$\Delta = \{A/C, B/C\}$$

Part (a)

Theory	Default	Justification
$T_0 = \{A\}$	A/C	$T_0 \models A$ $T_0 \not\models \neg C$ $T_0 \not\models C$
$T_1 = \{A, C\}$	none	$T_1 \models C$ violates A/C $T_1 \not\models B$ violates B/C

Therefore $\{A\} \mathrel{|\!\sim}_\Delta C$.

Part (b)

Theory	Default	Justification
$T_0 = \{B\}$	B/C	$T_0 \models B$ $T_0 \not\models \neg C$ $T_0 \not\models C$
$T_1 = \{B, C\}$	none	$T_1 \not\models A$ violates A/C $T_1 \models C$ violates B/C

Therefore $\{B\} \mathrel{|\!\sim}_\Delta C$.

Part (c)

Theory	Default	Justification
$T_0 = \{A \vee B\}$	none	$T_0 \not\models A$ violates A/C $T_0 \not\models B$ violates B/C

Therefore $\{A \vee B\} \mathrel{|\!\not\sim}_\Delta C$.

Exercise 2

In part (c) of the previous problem, we showed that $\{A \vee B\} \not\mid\!\sim_{\Delta} C$ despite $\{A\} \mid\!\sim_{\Delta} C$ and $\{B\} \mid\!\sim_{\Delta} C$. However, if $T_0 \vDash A \vee B$, it is either the case that $T_0 \vDash A$ or $T_0 \vDash B$. From the Principle of Case Reasoning, since in both cases we can infer C, we can infer C from the premise $A \vee B$. If this principle is true, as common sense suggests, the logic covered in this unit is inadequate.

Exercise 3

$$
\begin{aligned}
\Delta = \{&\text{HasFunding/Admit},\\
&\text{MeetsGPARequirements/Admit},\\
&\text{ResearchExperience/Admit}\}\\
\text{HasFunding/Admit} &> \text{ResearchExperience/Admit}\\
\text{ResearchExperience/Admit} &> \text{MeetsGPARequirements/Admit}
\end{aligned}
$$

For this strict partial ordering, I considered which factors a graduate admissions committee prioritizes when deciding whether or not to admit a student. If $T = \{\text{HasFunding}, \text{MeetsGPARequirements}, \text{ResearchExperience}, \neg\text{Admit}\}$, none of the defaults are applicable since $T \vDash \neg\text{Admit}$. The set of skeptical nonmonotonic consequences is then $\text{Cn}(T)$.